# Set Theory and Mathematical Reasoning

## Exercises

Page numbers refer to the full notes, `settheory.pdf`.

## 1 Sets and numbers

**Exercise 1.1** (p. 6) For every set, there is at least one correspondence between it and itself.

**Exercise 1.2** (p. 6) For a set with two elements there are precisely two correspondences between it and itself. For a set with three elements there are precisely six correspondences between it and itself.

**Exercise 1.3** (p. 6) If there is a correspondence between $A$ and $B$, and a correspondence between $B$ and $C$, then there is a correspondence between $A$ and $C$.

**Exercise 1.4** (p. 6) If there is a correspondence between $A$ and $B$, then there is a correspondence between $B$ and $A$.

**Exercise 1.5** (p. 7) Counting a set establishes a correspondence between it and the natural numbers below its cardinality.

**Exercise 1.6** (p. 7) If you can count two sets, and get the same cardinal number, then there is a correspondence between the two sets.

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## 2 Well-orderings, countable vs uncountable infinities

**Exercise 2.1** (p. 12) Show that any relation that is transitive (if $x$ is before $y$ and $y$ is before $z$, then $x$ is before $z$) and irreflexive (there is no $x$ with $x$ coming before $x$) must be antisymmetric (if $x$ is before $y$ then $y$ is *not* before $x$).

**Exercise 2.2** (p. 12) Show that the lexicographic ordering on ordered pairs of natural numbers is a well-ordering.

**Exercise 2.3** (p. 13) What is $(\omega + 1) + (\omega + 3)$? What is $(\omega + 2) + (\omega + 2)$? What is $(\omega + 3) + (\omega + 3)$? What is $(\omega + 6) + \omega$? Which, if any, of these are equal?

**Exercise 2.4** (p. 14) What is $2 \cdot (\omega + 3)$? What is $(\omega + 3) \cdot 2$? Is either of them equal to $\omega + \omega + 3 + 3$?

**Exercise 2.5** (p. 17) Show that this same argument works for *any* set, not just the set of natural numbers. That is, let $f$ be a function from $S$ to $\mathcal{P}(S)$, let $D = \{x \in S \mid x \notin f(x)\}$, and show that $D$ isn't in the range of $f$.

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## 3 Many countably infinite sets

**Exercise 3.1** (p. 19) Any infinite subset of a countably infinite set is countably infinite.

**Exercise 3.2** (p. 19) The union of two countably infinite sets is countably infinite.

**Exercise 3.3** (p. 19) The union of countably many sets that are each countably infinite is countably infinite.

**Exercise 3.4** (p. 19) The set of all finite sequences of natural numbers is countably infinite.

**Exercise 3.5** (p. 20) The set of all grammatical sentences in English is countably infinite.

**Exercise 3.6** (p. 20) The set of all syntactically correct computer programs is countably infinite.

**Exercise 3.7** (p. 20) The set of all definitions of sets of natural numbers is countably infinite.

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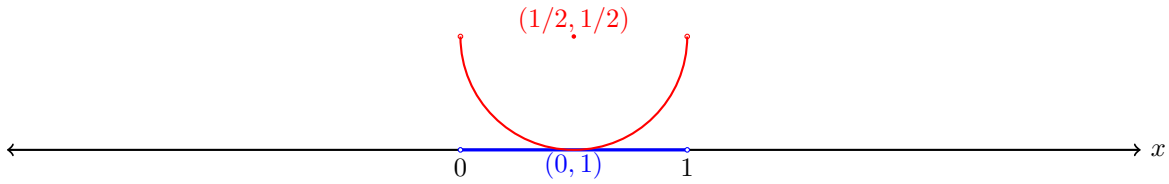
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## 4 Comparing the real numbers and the power set of the natural numbers

**Exercise 4.1** (p. 22) Show, using just these axioms, and the ones for ordering, that if $x_1 \circ y_1 = x_2 \circ y_2$, and $x_1 < x_2$, then $y_1 > y_2$.

**Exercise 4.2** (p. 25) Consider the sequence of nested intervals $[n, n+1]$, $[n.d_1, n.d_1 + .1]$, $[n.d_1d_2, n.d_1d_2 + .01]$, .... This is the set of all quantities that share a given $(n+1)$th approximation. Use the Continuity principle to show that there must be *at least* one quantity contained in the intersection of all these intervals, and use the Archimedean axiom to show that there must be *at most* one quantity contained in the intersection of all these intervals. (Hint: If there were two distinct quantities contained in this intersection, see what wold happen if you count how many copies of their difference go into the original unit.)

**Exercise 4.3** (p. 27) Show that there is a correspondence between the set $\{x \in \mathbb{R} \mid 0 < x < 1\}$ (all real numbers between 0 and 1, blue in the diagram below) and the set of points on the semicircle of radius 1/2, centered at $(1/2, 1/2)$ (red in the diagram below), not including the endpoints of either the interval or the semicircle. Make sure to show that all four properties of a correspondence hold.




**Exercise 4.4** (p. 27) Show that there is a correspondence between the set of points on the semicircle of radius 1/2, centered at $(1/2, 1/2)$ (not including the endpoints) and the set of *all* real numbers. Make sure to show that all four properties of a correspondence hold.

**Exercise 4.5** (p. 28) Show that there is a correspondence between the set of points in any particular finite interval of real numbers (not including endpoints) and the set of *all* real numbers.

**Exercise 4.6** (p. 29) Using binary notation, show that there is a function from the real numbers in the interval from 0 to 1 (including endpoints) to $\mathcal{P}(\mathbb{N})$, the power set of the natural numbers, that is total, and one-to-one, but *not* onto. (Remember that in binary notation, $0.{}_2011111\cdots = 0.{}_21$ for the same reason that $0.09999\cdots = 0.1$ in decimal notation.)

**Exercise 4.7** (p. 29) Using notation in any base greater than 2, show that there is a function from $\mathcal{P}(\mathbb{N})$ to the set of real numbers in the interval from 0 to 1 (including endpoints), that is total, and one-to-one, but *not* onto.

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## 5 The Axiom of Choice, the paradoxes

**Exercise 5.1** (p. 31) Show (using the Axiom of Choice) that every infinite set has a subset that has a correspondence with the natural numbers.

**Exercise 5.2** (p. 32) Show that for any $n$, these (i.e., first player guesses white if they see an odd number of white hats; first player guesses black if they see an odd number of white hats) are the *only* two strategies that can guarantee that everyone but the first player guesses correctly.

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## 6 The formal language, and the "small" axioms: Extensionality, Pairing, Union, and Empty Set

**Exercise 6.1** (p. 34) If $x \subseteq y$ and $y \subseteq z$ then $x \subseteq z$.

**Exercise 6.2** (p. 35) $\langle x, y\rangle = \langle z, w\rangle$ iff $x = z$ and $y = w$.

**Exercise 6.3** (p. 36) Using these definitions, show that each $n$ is the set of all natural numbers strictly less than $n$. (This will be a proof by induction. The base case is to show that this is true when $n = 0$, where there is no natural number less than $n$, and the induction step involves showing that if it's true for $n$, then it's also true for $n + 1$.)

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## 7 The "medium" axioms: Power set, (Foundation), and Separation

**Exercise 7.1** (p. 37) Find $\mathcal{P}(\emptyset)$, $\mathcal{P}(\mathcal{P}(\emptyset))$, and $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))$ and write them explicitly using curly braces.

**Exercise 7.2** (p. 37) If $x \subseteq y$ then $\mathcal{P}(x) \subseteq \mathcal{P}(y)$.

**Exercise 7.3** (p. 37) $x = \bigcup \mathcal{P}(x)$ and $x \subseteq \mathcal{P}(\bigcup x)$, but there exists a set $x$ such that $x \neq \mathcal{P}(\bigcup x)$.

**Exercise 7.4** (p. 38) Use the Separation axiom to prove that the empty set exists, using only the assumption that at least one set exists.

**Exercise 7.5** (p. 38) Use the Separation axiom to prove that there is no set of all sets. (Hint: Use Russell's paradox, letting $\varphi(w)$ be "$\neg(w \in w)$".)

**Exercise 7.6** (p. 39) If $R$ is a relation from $X$ to $Y$, and $S$ is a relation from $Y$ to $Z$, and $R$ and $S$ are both (functions/total/one-to-one/onto) then so is $R \circ S$ as a relation from $X$ to $Z$.

**Exercise 7.7** (p. 39) (Cantor's Theorem) Show that for any set $X$, there is no relation with domain $X$ and range $\mathcal{P}(X)$ that is both a function and onto.

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## 8 Infinity, formalized ordinals, and the well-ordering principle

**Exercise 8.1** (p. 40) No natural number has a correspondence to a proper subset of itself. (Hint: The best way to prove this in the formal axiomatic context is to show that the set of natural numbers that don't have a correspondence to a proper subset is inductive.)

**Exercise 8.2** (p. 40) $\omega$ does have a correspondence to a proper subset of itself.

**Exercise 8.3** (p. 41) Use a similar sequence of operations to show that $S(S(0)) + (S(0) + S(0)) = S(S(S(S(0))))$ (that is, that $2 + (1 + 1) = 4$).

**Exercise 8.4** (Commutativity of addition; p. 42) For all natural numbers $m, n$, $m + n = n + m$.

**Exercise 8.5** (Associativity of addition; p. 42) For all natural numbers $\ell, m, n$, $\ell + (m + n) = (\ell + m) + n$.

**Exercise 8.6** (p. 43) Prove (by induction on $n$) that the combinatorial definition of multiplication and the recursive definition agree for all $m, n$. (The base case will be very straightforward; for the induction step, there is a combinatorial definition of $m \cdot S(n)$, and there is a combinatorial definition of $m \cdot n + m$, and you will want to demonstrate a correspondence between the relevant sets in this combinatorial definition.)

**Exercise 8.7** (p. 44) Show that for all natural numbers $m, n$, $m \cdot n = n \cdot m$.

**Exercise 8.8** (p. 44) Show that for all natural numbers $\ell, m, n$, $\ell \cdot (m \cdot n) = (\ell \cdot m) \cdot n$.

**Exercise 8.9** (p. 44) Show, using the combinatorial definitions, that $m^2 = m \cdot m$. (This involves finding a correspondence between things like $f = \{\langle 0, k\rangle, \langle 1, \ell\rangle\}$ and ordered pairs of elements of $m$.)

**Exercise 8.10** (p. 44) Show, using the combinatorial definition, that the power set of $m$ has $2^m$ members. (This involves finding a correspondence between functions from $m$ to $\{0, 1\}$ and subsets of $m$.)

**Exercise 8.11** (p. 44) Show, by induction, that the recursive and combinatorial definitions of exponentiation agree.